Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

1. TEAM AND PROJECT OVERVIEW

Team Name

Calador

Project / Solution Name

Calador - AI Career & Skills Navigator (Future-Ready Workforce domain)

Name (Leader)

Soon Yen Ling

Email & Phone (Leader)

soonyling5@gmail.com & 0177281683

University Affiliation(s)

TARUMT , UTAR

Live Prototype URL

https://calador-sdc-hack.replit.app

Code Repository URL (public / access granted to judges)

https://github.com/sssoooonn/Calador-SDC-Hack

2. ARCHITECTURE & TECHNOLOGY STACK

2.1. System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Calador is a single-process Streamlit web app written in Python. It has no separate backend, database or AI model. The Streamlit interface collects a user's current role, skills, learning preferences and an optional CV (PDF/DOCX/TXT), or a team/student CSV. A rule-based engine in the same app detects skills by keyword matching and scores roles built from the Kaggle AI Jobs Market CSV (loaded with pandas): 90% skill coverage plus 10% background relevance. Trend and SDG preference weights only affect ranking. Two public feeds add live signals: the arXiv cs.AI RSS feed (research heat) and the Arbeitnow job-board API (live skill demand). Responses are cached for one hour and saved to local JSON files (trends_cache.json, live_jobs_cache.json), which serve as fallback when offline or in low-bandwidth mode. Outputs (role matches, skills radar, SDG mapping, free-resource reskilling plan, fairness check, team heatmaps) are shown with Altair charts and tables in English, Bahasa Malaysia or Chinese. Uploaded data stays in session memory. The app is hosted on Replit.

2. ARCHITECTURE & TECHNOLOGY STACK

2.2. Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.11.9 (incl. standard library: json, re, urllib)	Core language; networking, parsing and caching	PSF License
Streamlit 1.62.0; Altair 6.2.2	Web app framework and interface; charts (radar bars, heatmap, distributions)	Apache-2.0; BSD-3-Clause
pandas 3.0.1; NumPy 2.4.3	Loading and aggregating the job dataset, tables; numeric calculations (scores, trends)	BSD-3-Clause
feedparser 6.0.14	Reading the arXiv cs.AI RSS feed	BSD-2-Clause
pypdf 6.21.2; python-docx 1.2.0	Extracting text from uploaded PDF and DOCX CVs	BSD-3-Clause; MIT
Barlow and Oswald fonts (Google Fonts)	Interface typography (web fonts)	SIL Open Font License 1.1

3. APIs & CLOUD SERVICES

3.1. External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API and Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
arXiv RSS feed / arXiv API (arXiv.org, Cornell University)	Fetch the latest cs.AI paper titles to compute research-heat trend signals	Free (no key)
Arbeitnow Job Board API (Arbeitnow)	Fetch recent job ads to compute live skill-demand share	Free (no key)
Google Fonts (Google)	Load interface fonts in the browser (skipped in low-bandwidth mode)	Free

3. APIs & CLOUD SERVICES

3.2. Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit (Replit, Inc.)	Hosting and deployment of the Streamlit prototype	Sponsor-provided credits; region not specified

4. AI TOOLS, MODELS & DATASETS

4.1. AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
None	N/A	N/A - no AI/ML or LLM model is used; matching, trends, SDG mapping and plans are rule-based	N/A

4. AI TOOLS, MODELS & DATASETS

4.2. Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Sample team/student members	Created by our team in the app and CSV template	Synthetic	Own work	N
Free learning-resource links	Links only (no content copied): Kaggle Learn, Google MLCC, Hugging Face, fast.ai, Khan Academy, DeepLearning.AI, Elements of AI, Coursera, Docker, Pro Git, HRD Corp	Public	Each provider's terms	N
AI Jobs Market 2025-2026 (Salaries)	Kaggle (alitaqishah): kaggle.com/datasets/alitaqishah/ai-jobs-market-2025-2026-salaries	Public	As stated on the Kaggle dataset page	N
arXiv cs.AI latest paper titles	rss.arxiv.org/rss/cs.AI	Public (live feed)	arXiv terms of use	N
Arbeitnow job ads	arbeitnow.com/api/job-board-api	Public (live API)	Arbeitnow API terms	N

4. AI TOOLS, MODELS & DATASETS

4.3. AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	Used For	Extent of Use (minor / moderate / major)
Google Gemini	Setting up the development environment, code writing, and design ideas	Minor
Claude Sonnet 5.5 (Anthropic)	Code writing, debugging, and solving coding problems	Major
Replit	Code writing, debugging, and deployment	Major

4. AI TOOLS, MODELS & DATASETS

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Our team reviewed AI-generated code and suggestions to ensure they matched the project requirements. We checked the code, tested the functions, and debugged errors before using the output in the project. We also reviewed AI-generated designs and translations to ensure they were suitable and accurate. AI-generated content was not accepted without checking. Our team made the final decisions, corrected mistakes when necessary, and took responsibility for the quality, functionality, and accuracy of the final project.

5. SPONSOR-PROVIDED RESOURCES

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
Replit	Replit credits (platform)	Deploying the live prototype, debugging errors, and solving code problems

5. SPONSOR-PROVIDED RESOURCES

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Yes, Replit was very important to our project. We used Replit credits to deploy the live prototype, so judges can open it through a public link. We also used Replit throughout development to debug errors, solve code problems, and test the app after each change. This saved us a great deal of time and helped us finish the prototype before the deadline. The app itself is a standard Python Streamlit program, so it could be moved to another host, but this would have taken extra time and effort. The prototype does not depend on any sponsor generative-AI API.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.1. Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The app has no user accounts and no database. Information entered by users (role, skills, questionnaire answers, an optional CV, or a team CSV/table) is processed in memory during the Streamlit session and is not written to disk or kept afterwards. No user data is sent to any third party or AI API; the only outbound requests fetch public data (arXiv feed, Arbeitnow job ads) and contain no user input. The local cache files hold public data only, and low-bandwidth mode makes no external fetches. The interface tells users to upload only synthetic or sample CVs, and example members are synthetic. The app does not collect age, gender or ethnicity. Loading Google Fonts in the browser may reveal the visitor's IP address to Google (skipped in low-bandwidth mode). The app is hosted on Replit, whose standard server logs apply.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.2. Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

No generative AI is used, so hallucination is not a risk. Results come from rules and public data, and missing data is shown as unavailable instead of invented. The dataset is small, with about 20-50 records per role per year, and 2026 covers only January to March, so trends are signals rather than forecasts. Keyword matching only recognises skills in the dataset vocabulary. The arXiv research heat is not hiring demand, and the Arbeitnow data is mostly European rather than Malaysian. The weights and scenario multipliers are adjustable assumptions. Background relevance could add bias, so we capped it at 10% and added a skills-only mode and a fairness check, which cannot prove the dataset is unbiased. The app relies on external feeds and static course links, so we added cached fallback data and "no matching resource" messages. The results support human decisions and do not replace them.

7. COMPLIANCE & READINESS

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. DECLARATION

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature] *Soon*

[Date] 24/10/2026